KUMARAGURU COLLEGE OF TECHNOLOGY

Department of Artificial Intelligence and Data Science

DATA SCIENCE AND VISUALIZATION (24ADI204)

WEEK 6 REPORT – FEATURE ENGINEERING

Normalization, Scaling, Encoding and Skewness Correction

Youth Smoking and Drug Abuse Analysis

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Team 16

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1. Introduction

This report presents the Week 6 work of the project "Youth Smoking and Drug Abuse Analysis". The focus of this week is Feature Engineering, where the cleaned dataset is transformed into suitable formats for further analysis. The work includes identifying numerical and categorical variables, encoding categorical variables, applying standardization and Min-Max scaling, and examining skewness in the personality features.

The notebook uses a separate copy of the cleaned dataset (df_fe) so that the original cleaned dataset remains unchanged. The dataset contains 1,885 records and 32 columns.

2. Objective

To prepare the cleaned dataset for further analysis by applying suitable feature engineering techniques, including categorical encoding, normalization and scaling, and skewness analysis and correction where required.

3. Activities Performed

Identified numerical personality features and categorical drug-consumption features.

4. Approach and Data Preparation

4.1 Creating a Separate Feature Engineering Dataset

A separate copy of the cleaned dataset was created using df_fe = df_clean.copy(). All encoding and scaling operations were carried out on separate copies. This preserved the original cleaned dataset and allowed the different transformations to be compared.

4.2 Identification of Numerical and Categorical Features

The seven personality score columns were treated as numerical features. The 19 drug-consumption columns were treated as categorical features because their values are stored as consumption-level labels from CL0 to CL6.

Feature Type	Columns	Number of Features
Numerical	Nscore, Escore, Oscore, Ascore, Cscore, Impulsive, SS	7
Categorical	Alcohol, Amphet, Amyl, Benzos, Caff, Cannabis, Choc, Coke, Crack, Ecstasy,	19

	Heroin, Ketamine, Legalh, LSD, Meth, Mushrooms, Nicotine, Semer, VSA

Table 1: Identification of numerical and categorical features

5. Encoding Categorical Variables

5.1 Label Encoding

Label Encoding converts categorical labels into numerical values. In this dataset, the drug-consumption categories have a natural order from CL0 (never used) to CL6 (used in the last day). The notebook mapped these labels to integers 0 to 6 for all 19 drug-consumption columns.

Original Category	Encoded Value	Meaning
CL0	0	Never used
CL1	1	Used over a year ago
CL2	2	Used in the last year
CL3	3	Used in the last month
CL4	4	Used in the last week
CL5	5	Used in the last 24 hours
CL6	6	Used in the last day

Table 2: Label encoding map used for drug-consumption levels

The label-encoded dataset retained the same shape as the original dataset: 1,885 rows and 32 columns. The encoding changed the representation of the drug columns without adding new columns.

5.2 One-Hot Encoding

One-Hot Encoding represents a category using separate binary columns. The encoded Gender values were first mapped to Male and Female. The Gender column was then replaced with Gender_Label_Female and Gender_Label_Male. A value of 1 indicates the corresponding category, while 0 indicates that it does not apply.

Dataset	Rows	Columns
Original feature engineering dataset	1,885	32
After One-Hot Encoding	1,885	33

Table 3: Dataset shape after One-Hot Encoding

The number of columns increased from 32 to 33 because one Gender column was replaced by two binary columns.

6. Normalization and Scaling

6.1 Standardization

Standardization transforms numerical features so that they have approximately zero mean and unit standard deviation. StandardScaler was applied to the seven personality score columns: Nscore, Escore, Oscore, Ascore, Cscore, Impulsive and SS.

The notebook output confirms that the mean of each transformed feature is approximately 0.00 and the standard deviation is approximately 1.00. This puts the personality features on a comparable scale.

Feature	Mean after standardization	Standard deviation
Nscore	0.00 (approx.)	1.00

Escore	0.00 (approx.)	1.00
Oscore	0.00 (approx.)	1.00
Ascore	0.00 (approx.)	1.00
Cscore	0.00 (approx.)	1.00
Impulsive	0.00 (approx.)	1.00
SS	0.00 (approx.)	1.00

Table 4: Standardization check for personality features

6.2 Min-Max Scaling (Normalization)

Min-Max Scaling transforms numerical values into a fixed range from 0 to 1. MinMaxScaler was applied to the same seven personality score columns. The notebook displayed the transformed values, which fall within the 0 to 1 range. This method is useful when features need to be represented on a common bounded scale.

Standardization and Min-Max Scaling were performed on separate copies of the original feature engineering dataset. Therefore, the two methods can be compared without one transformation overwriting the other.

7. Skewness Analysis and Correction

7.1 Skewness Analysis

Skewness measures the asymmetry of a data distribution. Positive skewness indicates a longer right tail, negative skewness indicates a longer left tail, and a value close to zero indicates a relatively symmetrical distribution. Skewness was calculated for all seven personality features.

Personality Feature	Skewness
Nscore	-0.005123
Escore	-0.001738
Oscore	-0.010308
Ascore	-0.003897
Cscore	-0.005083
Impulsive	0.072795
SS	-0.040660

Table 5: Skewness values of personality features before transformation

All seven skewness values are close to zero. The highest absolute value is 0.072795 for Impulsive, which is still below the threshold of 0.5 used in the notebook. This indicates that the selected personality features do not show substantial skewness.

7.2 Yeo-Johnson Transformation Check

The Yeo-Johnson PowerTransformer was selected as the skewness correction method because it can handle both positive and negative values. The notebook selected features for transformation only when their absolute skewness was greater than 0.5.

No feature met this condition. Therefore, the selected skewed feature list was empty and Yeo-Johnson transformation was not applied to any column. The before-and-after skewness comparison remained unchanged. This is an appropriate outcome because the measured personality features were already approximately symmetrical; unnecessary transformation was avoided.

Check	Notebook result

Skewness threshold	Absolute skewness > 0.5
Features selected	None (empty list)
Yeo-Johnson transformation	Not applied
Before-and-after skewness	Values remained unchanged

Table 6: Result of the skewness correction check

8. Key Findings

• The cleaned dataset contained 1,885 records and 32 columns when feature engineering started.

• Seven numerical personality features and 19 categorical drug-consumption features were identified.

• Label Encoding converted the ordered drug-consumption levels CL0 to CL6 into values 0 to 6.

• One-Hot Encoding represented Gender using two binary columns, increasing the dataset width to 33 columns.

• Standardization produced approximately zero means and unit standard deviations for all seven personality features.

• Min-Max Scaling transformed the personality features to values between 0 and 1.

• All personality-feature skewness values were close to zero; no feature crossed the 0.5 threshold, so Yeo-Johnson transformation was not required.

9. Final Dataset Summary

Dataset Version	Shape (Rows × Columns)	Operation
df_fe	1,885 × 32	Separate copy of cleaned dataset
df_label_encoded	1,885 × 32	Label Encoding of 19 drug columns
df_onehot	1,885 × 33	One-Hot Encoding of Gender
df_standard	1,885 × 32	Standardization of 7 personality features
df_minmax	1,885 × 32	Min-Max Scaling of 7 personality features
df_transformed	1,885 × 32	No columns transformed; skewness threshold not met

Table 7: Summary of datasets created during feature engineering

The transformations were performed on separate dataset copies. The different shapes shown above reflect the additional binary Gender column in the One-Hot Encoded version; the other transformations changed feature values without changing the number of rows or columns.

10. Conclusion

Feature Engineering was carried out on the cleaned Youth Smoking and Drug Abuse dataset. Numerical and categorical features were identified, the ordered drug-consumption categories were label encoded, and Gender was one-hot encoded. Standardization and Min-Max Scaling were applied to the seven personality score features using separate dataset copies. Skewness analysis showed that all seven personality features were already close to symmetrical, so no feature required Yeo-Johnson transformation under the selected threshold. These steps prepared the dataset in different forms for subsequent analysis and modelling.